Measuring Instrument – Digital Multimeter

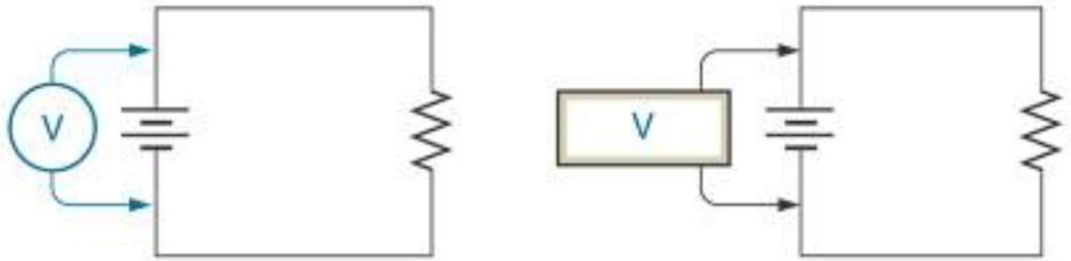
***** Multimeter***

- *Ammeter – current*
- *Voltmeter – voltage*
- *Ohmmeter – resistance*
- *Continuity Test*



Measuring Voltage

- The **instrument** is called a **<u>Voltmeter</u>**
  - **Connect** meter **<u>across</u>** component

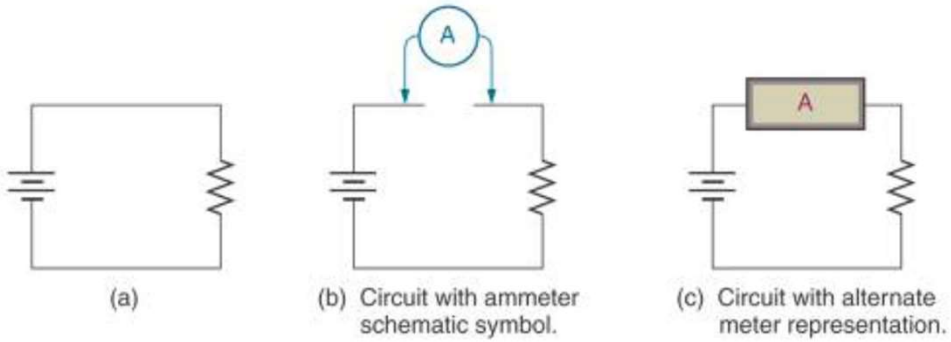


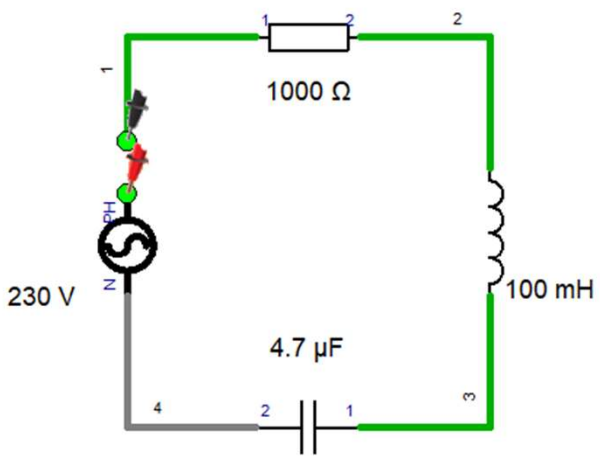
(a) Circuit with voltmeter schematic symbol.

(b) Circuit with alternate meter representation.

Measuring Current

- The **instrument** is called an **Ammeter**
- **Insert** the Ammeter in the current path







Automation Studio: Multimeter in AC current measurement mode

Measuring Resistance

- The instrument is called an **Ohm-meter.**
- The Multimeter measures resistance by sending a small current out through the test leads to the circuit under test.

- **Note:** Resistance should not be measured with power applied to the circuit.
- Usually, it is best to remove resistor from the circuit before measuring; as it can be affected by other components.

Continuity Check

- Continuity is the presence of a complete path for current flow.

Beep sound → Path complete (continuity present).
No beep → Path open (no continuity).

